Plasma-driven bubble generation at IR wavelengths has been demonstrated for 45-fs, 800-nm laser pulses [346,351,352] but the results of Fig. 15(c) suggest that it will be possible also at wavelengths > 1000 nm, which are often employed in LAL, and at longer pulse durations. The boundary of the (τ_L, λ_L, E_L) parameter space, in which they play a role, remains to be explored. It will be influenced by the temporal structure of the laser pulses. Intensity spikes characteristic for mlm Nd:YAG laser pulses facilitate plasma ignition in field-enhanced regions, and the threshold irradiance averaged over the spikes of mlm pulses is lower than for temporally smooth slm pulses (Fig. 4). Therefore, bubble formation at nanoparticles could likely be inhibited by using IR slm ns pulses – but once breakdown occurs, bubbles are larger (Fig. 5).

Laser-induced particle fragmentation. Nanoparticle generation by LAL often leads to an undesirably broad size distribution. Secondary laser irradiation can be used to fragment large free nanoparticles for homogenization of their size and modification of their crystalline structure [95,96,98,100]. Atomistic simulations and X-ray scattering methods have provided deep insight in the fragmentation mechanisms [97,99,353,354]. The simulations assume energy deposition into the particle, which is typical for laser wavelengths near the plasmon resonance. Stress-confined energy deposition by ultrashort laser pulses favors the generation of thermoelastic tensile and (in nonspherical particles) shear stress, which in combination with heat-diffusion-mediated bubble formation results in fragmentation. Rapid cooling yields a crystalline structure boosting the reactivity and catalytic efficiency of the fragments.

The use of off-resonance IR laser pulses with plasma-mediated bubble generation around the nanoparticle modulates the interplay between internal thermoelastic stress generation and external forces. At low radiant exposures close to the bubble threshold, nanoparticle integrity is well maintained by the relatively small temperature rise inside the particle that avoids melting. Targeted nanosurgeries on cells such as photoporation for gene modification or gene silencing with antibody conjugated nanoparticles [25,355] can be performed with good transfection efficiency and nanoparticle integrity [347,356]. However, at large radiant exposure values, where events become more vigorous, the high plasma pressure in the field-enhanced region and the high pressure at bubble collapse may induce fragmentation, or at least support the action of the internal thermoelastic stress. The concentration of plasma pressure in two field-enhanced regions [Fig. 15(a)] leads to shear stresses that are more favorable for fragmentation than homogeneous compressive stress. In particle clouds, the interaction between adjacent bubbles leads to aspherical collapse events with liquid jet impact, which also produces shear stress [63,74,357]. Hence, tuning the laser wavelength could alter the fragmentation mechanism for metallic nanoparticles, which

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would affect also the physicochemical properties of the fragments. Nevertheless, fragmentation mediated through energy deposition into an absorbing particle remains the more universal mechanism. Here, the use of longer wavelengths can help to maintain its chemical integrity by reducing the occurrence of multiphoton-mediated alterations [342,358].

Plasma-mediated chemistry. Laser ablation and particle fragmentation in liquids are considered physical synthesis techniques but accompanied by chemical changes even if they are performed in water [72,75,359-361]. Many changes are either directly produced by free-electrons in the laser plasma or indirectly via free-electron-generated reactive oxygen species as reviewed in [98]. Hydrogen, O_2 and H_2O_2 are produced through ns breakdown at nanoparticles [362,363] and during LAL for metallic nanoparticle generation [339]. Direct free-electron-mediated disintegration of water dominates for noble metals such as gold and platinum, while redox reactions at the particle surface dominate for less noble metals like aluminum. They lead to massive gas evolution, which forms persistent bubbles disturbing the LAL process, as discussed further above. For gold and platinum, the plasma liquid chemistry provides a large H_2O_2 yield, with platinum being the best catalyst [339]. Laser-induced water splitting works also without nanoparticle support [364]. Yan et al. demonstrated efficient H_2 and H_2O_2 generation using energetic 532-nm ns pulses weakly focused in pure water, which yields a large interaction volume for plasma-mediated chemistry [365]. In high-density water plasmas, free-electron-mediated molecular disintegration goes along with thermal dissociation [172,362,366] and it is not easy to separate the contributions of both factors. Water splitting and other free-electron-mediated chemical reactions can also occur in cold plasmas created by series of fs pulses below the cavitation threshold (section 6.3). Their relative importance increases more and more with decreasing temperature. In pure water, they will dominate below ≈ 3000 K, above which thermal dissociation sets in [172].

A generalization of the combination of plasma-mediated and surface chemistry in LAL and laser fragmentation is reactive laser synthesis, in which different solvents or adding electrolytes/ligands allows control of the nanoparticles' surface chemistry and elemental composition [98,100]. The ability to produce nanoparticles with controlled surface ligands, and tunable composition has broadened the range of LAL-synthesized nanoparticle applications to catalysis [98,100], biotechnology[341], photovoltaics[367], and nanomedicine [368-370].

Application of multiple pulses in LIBS. In LIBS, plasma-mediated ablation is not the primary goal but a way for obtaining a characteristic spectroscopic signal from the atomized and excited target material after the initial continuum emission from the plasma has ceased [84,88,101-103]. The laser beam is

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usually scanned to explore different regions of the target, and pulse sequences with low or moderate repetition rate but large pulse energy are employed to maximize the spectroscopic signal. Therefore, shielding of the target by bubbles from previous pulses is of little concern but shielding by plasma formation in the liquid is a major problem. One could avoid this by adjusting the spot size such that the radiant exposure at the target remains close to the ablation threshold. However, this approach is difficult to realize in many practical applications. A way out is the application of double or even triple laser pulses during a time interval shorter than the cavitation bubble oscillation time. Application of double pulses has been introduced already in the early work on LIBS in liquids [82] and is a topic of active research [87,371-374]. A sequence of more than two pulses (pre-pulse for void creation and up to 6 ablation pulses) has been used to improve the efficiency of pulsed laser tissue ablation in confined liquid environment while reducing mechanical side effects [375]. For LIBS in a liquid environment, triple pulses could be particularly useful: The cavity created by the first pulse frees the optical path through the target. An energetic second pulse can then ablate target material without shielding by plasma formation in the liquid. A third pulse further atomizes and excites the ablated material inside the bubble to improve the spectroscopic signal. For good coupling of laser energy into the bubble, pulses 2 and 3 should be applied when the wall of the expanding bubble is still smooth. Separation of the three tasks into three steps with appropriate timing and pulse energy could provide an even better overall performance than double pulses.

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